

Symmetry (S) Regime Framework — Working Proposal

1. Core Metric: Symmetry (S)

Definition

$$S = \frac{|n_{above} - n_{below}|}{n_{above} + n_{below}}$$

Meaning

Measures **directional imbalance inside a rolling window**.

- Low S → balanced buying/selling activity
 - High S → one-sided dominance
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2. Problem with raw S

Raw S alone answers only:

“Is there imbalance?”

It does NOT answer:

“Is this imbalance normal or meaningful in context?”

So raw S is **not enough for regime detection**.

3. Proposed Extension: Contextual Symmetry (S_z)

Step 1: Rolling baseline

$$S_{mean}, S_{std}$$

Step 2: Standardization

$$S_z = \frac{S - S_{mean}}{S_{std}}$$

4. What S_z adds

Instead of absolute imbalance, S_z measures:

“How unusual is current imbalance relative to recent history?”

So it converts:

- raw signal → **context-aware signal**
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5. Proposed Regime States

Based on S_z :

Balanced Regime

- $S_z < -0.5$
 - symmetry unusually low
 - stable / equilibrium-like behavior
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Rising Pressure Regime

- $-0.5 \leq S_z \leq 1.0$
 - imbalance increasing but not extreme
 - early directional formation
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Dominance Regime

- $S_z > 1.0$
 - unusually high imbalance
 - strong directional pressure / breakout conditions
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6. Conceptual Interpretation

This framework separates:

Raw measurement

- $S \rightarrow$ “what is happening now”

Contextual meaning

- $S_z \rightarrow$ “how unusual is it”

Behavioral classification

- regimes → “what kind of state the system is in”
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7. What this framework is trying to achieve

It is moving toward a system that can:

detect **market state transitions**, not just measure indicators

Specifically:

- equilibrium (balanced noise)
 - early imbalance formation
 - dominance / breakout conditions
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8. Important caution (why you paused correctly)

This introduces:

- thresholds
- normalization logic
- regime definitions

So now the system shifts from:

descriptive metrics → decision framework

That’s a conceptual jump, and it’s good you paused here.

9. Current status of your system

You already have:

- $C \rightarrow$ structural distortion
- scale $\rightarrow$ energy baseline
- imbalance $\rightarrow$ directional bias
- reversion $\rightarrow$ equilibrium force
- $S \rightarrow$ directional symmetry

Now you are *considering*:

- $S_z \rightarrow$ contextual symmetry
 - regime classification layer (not implemented yet)
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10. Key takeaway

What was proposed is not new math—it is:

turning S from a metric into a **state classifier** using **historical context**